Yuqi Gu

(Updated on November 3, 2026)

Email: yuqi.gu@columbia.edu. **Website:** https://yuqigu.github.io/.

Address: Department of Statistics, Columbia University, Room 928 SSW, 1255 Amsterdam Avenue, New York, NY 10027, USA.

Employment

Columbia University, New York, NY, USA.
Assistant Professor (tenure-track), Department of Statistics. 07/2021 – Present
Member, Data Science Institute. 07/2022 – Present

Duke University, Durham, NC, USA.
Postdoctoral Researcher, Department of Statistical Science. 07/2020 – 06/2021

Education

University of Michigan, Ann Arbor, MI, USA. 08/2015 – 05/2020
Ph.D., Statistics.

Tsinghua University, Beijing, China. 09/2011 – 07/2015
B.S., Pure and Applied Mathematics.

Research Interests

My research develops statistical theory and methodology for uncovering **latent structure** in complex, high-dimensional data and AI systems. A unifying goal is to make latent structure and representation learning identifiable, computationally scalable, and statistically reliable, with principled uncertainty quantification and causal interpretation.

- **Identifiable deep generative models and causal representation learning:** Identifiability and structure discovery in deep generative models and deep latent variable models, including latent graphical and causal structures.
- **High-dimensional estimation and inference for latent structure:** Spectral, tensor, likelihood-based, and gradient-based methods for latent class, mixed-membership, and nonlinear low-rank models, with statistical optimality guarantees and uncertainty quantification.
- **Statistical measurement and AI evaluation:** Latent trait models, item response theory, cognitive diagnosis, and representation-assisted methods for measuring and comparing capabilities of LLMs and other AI systems, with broader applications to educational, psychological, and biomedical data.

Publications

Underlined are my student or postdoc advisees. ✉ indicates I am the corresponding author. * indicates co-first authors. † indicates alphabetical authorship.

Preprints in Submission:

42. From Good Starts to Optimal Inference: Generalized Latent Factor Models with Missingness and Implicit Regularization.
Chengzhu Huang and **Yuqi Gu**✉. (2026+) (preprint arXiv:2609.11740)
(This paper received the 2026 International Conference on Statistics and Data Science (ICSDS) Student Travel Award)

41. Mixed Membership Model of Low-rank Matrices with Multimodal Extension.
David Snider, Zhongyuan Lyu, Jian Kang, and **Yuqi Gu**✉. (2026+) (preprint arXiv:2608.18953)

40. Beyond Local Independence: High-Dimensional Latent Class Graphical Models with Shared Block Structure.
Seunghyun Lee and **Yuqi Gu**✉. (2026+) (preprint arXiv:2606.29631)

39. Scalable Variational Inference for Probabilistic Boolean Matrix Factorization with Unknown Latent Dimension.
Russell Kunes, Mingzhang Yin, and **Yuqi Gu**✉. (2026+) (preprint available here)

38. Scalable Covariate-Assisted Spectral Estimation of Grade-of-Membership Models.
Zhiyu Xu and **Yuqi Gu**✉. (2026+) (preprint arXiv:2601.17265)

Preprints Under Journal Revisions:

37. CRYSTAL: Cluster-geometry-inspired Spectral Algorithm for Binary Latent Feature Models.
Yuqi Gu†✉, Chengzhu Huang†, and Zhiyu Xu†. (2026+) (preprint available here)
Under "Major Revision" at *Journal of the American Statistical Association*.

36. Discrete Causal Representation Learning.
Wenjin Zhang, Yixin Wang, and **Yuqi Gu**✉. (2026+) (preprint arXiv:2603.25017)
(This paper received the *ASA SLDS Section 2026 Student Paper Award*)
Under "Major Revision" at *Journal of the American Statistical Association*.

35. Bayesian Latent Class Regression with Interpretable Binary Profiles.
Yuren Zhou, **Yuqi Gu**, David B. Dunson. (2026+) (preprint arXiv:2503.17531)
Under "Major Revision" at *Biometrika*.

34. Identifiable Bayesian Deep Generative Copulas with Unknown Layer Widths for Data with Arbitrary Marginal Distributions.
Joseph Feldman and **Yuqi Gu**. (2026+) (preprint arXiv:2605.27523)
Under "R&R" at *Journal of the American Statistical Association*.

33. Covariance Projected Spectral Clustering for High-Dimensional Anisotropic Mixtures: Practical Algorithm and Optimal Rates.
Chengzhu Huang and **Yuqi Gu**✉. (2026+) (preprint arXiv:2502.02580)
(This paper received the *2026 IMS Hannan Graduate Student Travel Award*)
Under "R&R" at *Journal of the Royal Statistical Society Series B: Statistical Methodology*.

32. Valid and Efficient Two-Stage Latent Subgroup Analysis with Observational Data.
Yuanhui Luo, Xinzhou Guo✉, and **Yuqi Gu**✉. (2026+) (preprint arXiv:2512.24223)
Under "R&R" at *Journal of the Royal Statistical Society Series B: Statistical Methodology*.

31. Scalable Text-Embedding-informed Cognitive Diagnosis of Large Language Models.
Jia Liu, Zhiyu Xu, and **Yuqi Gu**✉. (2026+) (preprint arXiv:2603.14676)
Revised and Resubmitted to *Psychometrika*.

Published/Accepted Peer-reviewed Journal Publications:

30. Deep Discrete Encoders: Identifiable Deep Generative Models for Rich Data with Discrete Latent Layers.
Seunghyun Lee and **Yuqi Gu**✉.
Journal of the American Statistical Association (2026), 121(553):194–208.
(This paper received the *ASA SLDS Section 2025 Student Paper Award*)

29. Adaptive Transfer Clustering: A Unified Framework.
Yuqi Gu†, Zhongyuan Lyu†, Kaizheng Wang†.
Journal of the American Statistical Association (2026+), accepted.

28. Generalized Grade-of-Membership Estimation for High-dimensional Locally Dependent Data.
Ling Chen*, Chengzhu Huang*, **Yuqi Gu**✉.
Journal of the American Statistical Association (2026+), accepted.

27. Bayesian Deep Generative Models for Multiplex Networks with Multiscale Overlapping Clusters. Yuren Zhou, **Yuqi Gu**, David B. Dunson.
Journal of the American Statistical Association (2026+), accepted.

26. Latency-Response Theory Model: Evaluating Large Language Models via Response Accuracy and Chain-of-Thought Length.
Zhiyu Xu, Jia Liu, Yixin Wang, and **Yuqi Gu**✉.
Annals of Applied Statistics (2026), 20(3):2253–2273.
(This paper received the *ASA SLDS Section 2026 Student Paper Award*)

25. On Theoretical Identifiability of Discrete Latent Causal Graphical Models.
Seunghyun Lee and **Yuqi Gu**. (2026).
Transactions on Machine Learning Research.

24. A Blockwise Mixed Membership Model for Multivariate Longitudinal Data: Discovering Clinical Heterogeneity and Identifying Parkinson's Disease Subtypes.
Kai Kang✉ and **Yuqi Gu**✉.
Annals of Applied Statistics (2026), 20(1):452–475.

23. Spectral Clustering with Likelihood Refinement for High-dimensional Latent Class Recovery.
Zhongyuan Lyu and **Yuqi Gu**✉.
Psychometrika (2026), 91:695–723.

22. Constructive Q-Matrix Identifiability via Novel Tensor Unfolding.
Yuqi Gu✉.
Psychometrika (2026), 91:131–150.

21. Deep Generative Modeling for Cognitive Diagnosis via Exploratory DeepCDMs.
Jia Liu and **Yuqi Gu**✉.
Psychometrika (2026), 91:151–176.

20. Degree-heterogeneous Latent Class Analysis for High-dimensional Discrete Data.
Zhongyuan Lyu, Ling Chen, **Yuqi Gu**✉.
Journal of the American Statistical Association (2025), 120(552), 2435–2448.

19. Learning from Similar Linear Representations: Adaptivity, Minimaxity, and Robustness.
Ye Tian, **Yuqi Gu**, Yang Feng.
Journal of Machine Learning Research (2025), 26(187):1–125.
(This paper received the *ASA SLDS Section 2025 Student Paper Award*)

18. Blessing of Dependence: Identifiability and Geometry of Discrete Models with Multiple Binary Latent Variables.
Yuqi Gu✉.
Bernoulli (2025), 31(2):948–972.

17. Exploratory General-response Cognitive Diagnostic Models with Higher-order Structures.
Jia Liu, Seunghyun Lee, **Yuqi Gu**✉.
Psychometrika (2025), 90:1651–1678.

16. Going Deep in Diagnostic Modeling: Deep Cognitive Diagnostic Models (DeepCDMs).
Yuqi Gu✉.
Psychometrika (2024), 89:118–150.

15. Latent Conjunctive Bayesian Network: Unify Attribute Hierarchy and Bayesian Network for Cognitive Diagnosis.
Seunghyun Lee and **Yuqi Gu**✉.
Annals of Applied Statistics (2024), 18(3):1988–2011.
(This paper received the *2024 IMS Hannan Graduate Student Travel Award*)

14. A Spectral Method for Identifiable Grade of Membership Analysis with Binary Responses.
Ling Chen and **Yuqi Gu**✉.
Psychometrika (2024), 89:626–657.

13. New Paradigm of Identifiable General-response Cognitive Diagnostic Models: Beyond Categorical Data.
Seunghyun Lee and **Yuqi Gu**✉.
Psychometrika (2024), 89:1304–1336.
(This paper received the *2023 Psychometric Society Student Travel Award*)

12. New directions in Algebraic Statistics: Three challenges from 2023.
Yulia Alexandr, Miles Bakenhus, Mark Curiel, Sameer Deshpande, Elizabeth Gross, **Yuqi Gu**, Max Hill, Joseph Johnson, Bryson Kagy, Vishesh Karwa, Jiayi Li, Hanbaek Lyu, Sonja Petrović and Jose Israel Rodriguez.
Algebraic Statistics (2024), 15(2):357–382.

11. Bayesian Pyramids: Identifiable Multilayer Discrete Latent Structure Models for Discrete Data.
Yuqi Gu✉ and David B. Dunson.
Journal of the Royal Statistical Society Series B: Statistical Methodology (2023), 85(2):399–426.

10. Generic Identifiability of the DINA Model and Blessing of Latent Dependence.
    Yuqi Gu✉.
    Psychometrika (2023), 88:117–131.

9. A Joint MLE Approach to Large-Scale Structured Latent Attribute Analysis.
    Yuqi Gu✉ and Gongjun Xu.
    Journal of the American Statistical Association (2023), 118(541):746–760.

8. Dimension-Grouped Mixed Membership Models for Multivariate Categorical Data.
    Yuqi Gu✉, Elena A. Erosheva, Gongjun Xu, and David B. Dunson.
    Journal of Machine Learning Research (2023), 24(88):1–49.

7. A Tensor-EM Method for Large-Scale Latent Class Analysis with Binary Responses.
    Zhenghao Zeng, **Yuqi Gu**, and Gongjun Xu.
    Psychometrika (2023), 88(2):580–612.

6. Identifiability of Hierarchical Latent Attribute Models.
    Yuqi Gu and Gongjun Xu.
    Statistica Sinica (2023), 33, 2561–2591.

5. Sufficient and Necessary Conditions for the Identifiability of the Q-matrix.
    Yuqi Gu and Gongjun Xu.
    Statistica Sinica (2021), 31, 449–472.

4. Partial Identifiability of Restricted Latent Class Models.
    Yuqi Gu and Gongjun Xu.
    Annals of Statistics (2020), 48(4):2082–2107.

3. Learning Attribute Patterns in High-Dimensional Structured Latent Attributes Models.
    Yuqi Gu and Gongjun Xu.
    Journal of Machine Learning Research (2019), 20(115):1–58.

2. The Sufficient and Necessary Condition for the Identifiability and Estimability of the DINA Model.
    Yuqi Gu and Gongjun Xu.
    Psychometrika (2019), 84(2):468–483.

1. Hypothesis Testing of the Q-Matrix.
    Yuqi Gu, Jingchen Liu, Gongjun Xu and Zhiliang Ying.
    Psychometrika (2018), 83(3):515–537.

Technical Report:

- Unfolding Tensors to Identify the Graph in Discrete Latent Bipartite Graphical Models.
  Yuqi Gu✉. (2025) (preprint arXiv:2501.10897)

Book Chapter:

- Chapter in the book draft *Bayesian Deep Learning*: "Interpretable and identifiable deep generative models via Bayesian pyramids and their extensions."
  Yuqi Gu*, Yuren Zhou*, David B. Dunson.

Published Discussions:

- Discussion of "Vintage Factor Analysis with Varimax Performs Statistical Inference" by Rohe & Zeng.
  Yinqiu He, **Yuqi Gu**, and Zhiliang Ying.
  Journal of the Royal Statistical Society Series B: Statistical Methodology (2023), 85(4):1071–1074.

Students and Postdoc Advisees

(Unless otherwise specified, the following advisees are from Columbia University.)

—— Postdoctoral Researcher ——

Current Postdoc:

- **Jia Liu**. Since 11/2025.

Former Postdoc:

- **Zhongyuan Lyu**. 07/2023 – 08/2025. (co-mentored with Kaizheng Wang)

  First job: Lecturer (Assistant Professor) in the Discipline of Business Analytics at the University of Sydney Business School.

—— Ph.D. Students ——

Graduated Ph.D. Students:

- **Seunghyun Lee**. Ph.D. graduated in May 2026. (co-advised with Sumit Mukherjee)

  Dissertation Title: Learning graphical models with intractable likelihoods.

  First job: Postdoctoral Researcher in the Department of Statistics at Harvard University.

- **Ling Chen**. Ph.D. graduated in May 2025. (co-advised with Jingchen Liu)

  Dissertation Title: Modern Latent Variable Analysis: High-dimensional Grade of Membership Models and Process Data.

  First job: Quantitative Researcher at Tudor Investment Corporation.

Current Ph.D. Students:

- Chengzhu Huang. Since 09/2023.
- Zhiyu Xu. Since 09/2024.
- David Snider. Since 09/2025.

- Jiatu Li. Since 09/2025.
- Wenjin Zhang. Since 09/2025.

—— Mentored Master's Student Researcher ——

- Junze Zhou. Since 09/2024.
  Placement: Biological and Artificial Intelligence (BioAI) PhD Program at Cold Spring Harbor Laboratory, Fall 2026.

—— Mentored Undergraduate Student Researchers ——

- Sihao Kong, USTC, 2026.
- Yifan Zhao, USTC, 2026.
- Wenjin Zhang, USTC, 2024.
  Placement: Statistics Ph.D. Program at Columbia University, since Fall 2025.
- Lehao Fu, USTC, 2023.
  Placement: Statistics Ph.D. Program at Cornell University, since Fall 2024.
- Zhiyu Xu, Tsinghua University, 2023.
  Placement: Statistics Ph.D. Program at Columbia University, since Fall 2024.
- Chengzhu Huang, USTC, 2022.
  Placement: Statistics Ph.D. Program at Columbia University, since Fall 2023.

—— Mentored Visiting Ph.D. Student ——

- Jia Liu, 11/2023 – 10/2024.
  Ph.D. student in Statistics from the Northeast Normal University, China.

—— Mentored High School Student Researcher ——

- Ari Stockwell, Riverdale Country School, Summer 2024.

  Mentored research project: Applying longitudinal mixed membership modeling to analyze election survey data to understand how the U.S. political ideology evolves.

  Placement: Admitted into Columbia College for Fall 2025.

Honors, Awards, and Grants

- Columbia University Lenfest Junior Faculty Development Grant, 2026
- Best Reviewer Award for *Psychometrika* from the Psychometric Society, 2024
- Outstanding Reviewer, American Educational Research Association and Journal of Educational and Behavioral Statistics, 2023
- National Science Foundation Grant DMS-2210796, "Latent Dependence and Identifiable, Graphical, Deep Modeling of Discrete Latent Variables", $150,000.
  Period: 09/01/2022 – 08/31/2026. Role: (Sole) Principal Investigator.

- IMS New Researcher Travel Award, Institute of Mathematical Statistics (IMS), 2021
- ProQuest Distinguished Dissertation Awards Honorable Mention, University of Michigan, 2020
- IMS Lawrence D. Brown Ph.D. Student Award, Institute of Mathematical Statistics, 2019
- Rackham Predoctoral Fellowship, Rackham Graduate School, UMich, 2019–2020
- Outstanding Graduate Student Instructor Award, Department of Statistics, UMich, 2019
- Tsinghua Xuetang Talents Program in Mathematics, Tsinghua University, 2013–2015

Student Paper Awards with Yuqi Gu as the Faculty Advisor and Senior Author

- 2026 IMS International Conference on Statistics and Data Science (ICSDS) Student Travel Award to Chengzhu Huang for our work on "From Good Starts to Optimal Inference: Generalized Latent Factor Models with Missingness and Implicit Regularization"
- 2026 ASA Statistical Learning and Data Science (SLDS) Student Paper Award to Zhiyu Xu for our work on "Latency Response Theory Model for Evaluating Large Language Models"
- 2026 ASA Statistical Learning and Data Science (SLDS) Student Paper Award to Wenjin Zhang for our work on "Discrete Causal Representation Learning"
- 2026 IMS Hannan Graduate Student Travel Award to Chengzhu Huang for our work on "Covariance Projected Spectral Clustering"
- 2025 ASA Statistical Learning and Data Science (SLDS) Student Paper Award and *Korean International Statistical Society Outstanding Student Paper Award* to Seunghyun Lee for our work on "Deep Discrete Encoders"
- 2024 IMS Hannan Graduate Student Travel Award to Seunghyun Lee for our work on "Latent Conjunctive Bayesian Network"
- 2023 Psychometric Society Student Travel Award to Seunghyun Lee for our work on "New paradigm of identifiable general-response cognitive diagnostic models"

Teaching Experience

As a Primary Instructor at Columbia University:

Statistics GR6102: Applied Statistics II. Spring 2026

Statistics GR6103: Applied Statistics III: Statistical Learning with Latent Structures. Fall 2025

Statistics GR6102: Applied Statistics II. Spring 2025

Statistics GR5205 / GU4205: Linear Regression Models. Fall 2024

Statistics GR6102: Applied Statistics II. Spring 2024

Statistics GR5205 / GU4205: Linear Regression Models. Fall 2023

Statistics GR6102: Applied Statistics II.	Spring 2023
Statistics GR5205 / GU4205: Linear Regression Models.	Fall 2022
Statistics GR6102: Applied Statistics II, co-instructed with Andrew Gelman.	Spring 2022
Statistics GR5205 / GU4205: Linear Regression Models.	Fall 2021

Invited Talks

2026:

- Invited Talk, 2026 IMS International Conference on Statistics and Data Science (ICSDS), Split, Croatia, 12/2026.
- Invited Talk, Seminar, Department of Statistics and Data Science, Washington University in St. Louis, 11/2026.
- Invited Talk, Conference on Statistical Foundations of Artificial Intelligence (CSFAI), University of California, Santa Barbara, 11/2026.
- Invited Talk, ASA Statistical Learning and Data Science (SLDS) Conference, New York City, 11/2026.
- Invited Talk, Seminar, Department of Biostatistics, Vanderbilt University, 10/2026.
- Invited Talk, Seminar, Department of Statistics and Data Science, Cornell University, 09/2026.
- Invited Talk, Seminar, Department of Biostatistics, Virginia Commonwealth University, 09/2026.
- Invited Talk, Seminar, Department of Epidemiology and Biostatistics, Memorial Sloan Kettering Cancer Center, New York City, 09/2026.
- Invited Talk, ICSA-Canada Chapter Symposium, Montreal, Canada, 08/2026.
- **Invited Speaker designated by the Psychometric Society President and Program Committee**, International Meeting of the Psychometric Society (IMPS) 2026, Seoul, Republic of Korea, 07/2026.
- Invited Talk, ICML 2026 Workshop "*Connecting Low-rank Representations in AI: From Practice to Theory*", Seoul, Republic of Korea, 07/2026.
- Invited Talk, 2026 ICSA China Conference, Shenzhen, China, 06/2026.
- Invited Talk, IMS Asia Pacific Rim Meeting, Hong Kong, 06/2026.
- Invited Talk, Network Stochastic Processes and Time Series (NeST) Online Seminar Series, 06/2026.
- Invited Talk, Statistical Inference for Network Models, NetSci 2026 Satellite Symposium, Boston, 06/2026.
- Invited Talk, Department of Statistics Student Seminar, Columbia University, 04/2026.

- Invited Talk, Seminar, Pennsylvania State University Statistics Colloquium, 03/2026.
- Invited Talk, Seminar, University of Virginia Statistics Colloquium, 03/2026.
- Invited Talk, Causal identification and discovery Workshop, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK, 03/2026.
- Invited Talk, Joint workshop of SNAB (Statistical Network Analysis and Beyond) and RMTA (random matrix theory and analysis), Nanjing, China, 01/2026.

2025:

- Invited Talk, Workshop at Imperial College London: "Statistics and Machine Learning for Networks and High-Dimensional Data", London, UK, 12/2025.
- Invited Talk, 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics), King's College London, 12/2025.
- Invited Talk, Seminar, Department of Statistics at London School of Economics and Political Science, 12/2025.
- Invited Talk, Seminar, Department of Statistics at University of Illinois Urbana-Champaign, 10/2025.
- Invited Talk, Big Data and Artificial Intelligence in Econometrics, Finance, and Statistics, University of Chicago, 10/2025.
- Invited Talk, 14th International Conference on Bayesian Nonparametrics (BNP 14), UCLA, 06/2025.
- Invited Talk, Workshop on Statistical Network Analysis and Beyond (SNAB 2025), Tokyo, Japan, 06/2025.
- Invited Talk, International Chinese Statistical Association (ICSA) 2025 Applied Statistics Symposium, Storrs, Connecticut, 06/2025.
- Invited Talk, Seminar, The University of Hong Kong (HKU) Business School, 05/2025.
- Invited Talk, Seminar on Statistics, Department of Mathematics, Hong Kong University of Science and Technology, 05/2025.
- Invited Talk, Math and Data (MaD) Seminar, Center for Data Science (CDS) at New York University, 04/2025.
- Invited Talk, Seminar, Department of Statistics, University of California, Irvine, 01/2025.
- Invited Talk, 2025 Statistics Empowering Data Science (SEEDS) Conference, Marshall School of Business, University of Southern California, 01/2025.

2024:

- Invited Talk, 2024 IMS International Conference on Statistics and Data Science (ICSDS), Nice, France, 12/2024.

- Invited Talk, Seminar, Department of Statistics, Rutgers University, Piscataway, New Jersey, 11/2024.
- Invited Talk, Seminar, Department of Statistics, University of South Carolina, Columbia, South Carolina, 11/2024.
- Invited Talk, Seminar, New York University (NYU) Center for Practice and Research at the Intersection of Information, Society, and Methodology (PRIISM), 10/2024.
- Invited Talk, Joint Statistical Meetings (JSM), Portland, Oregon, 08/2024.
- Invited Talk, Seminar, Department of Mathematical Sciences, Tsinghua University, Beijing, China, 06/2024.
- Invited Talk, 2024 Workshop on Statistical Network Analysis and Beyond (SNAB 2024), Nassau, Bahamas, 06/2024.
- **Invited Talk (Keynote)**, UC San Diego Halıcıoğlu Data Science Institute Causality Workshop, San Diego, California, 04/2024.
- Invited Talk, Workshop on Theory and Foundations of Statistics in the Era of Big Data, Florida State University, Tallahassee, Florida, 04/2024.
- Invited Talk, 2024 National Council for Measurement in Education (NCME) Annual Meeting, Philadelphia, Pennsylvania, 04/2024.
- Invited Talk, 2024 Eastern North American Region (ENAR) of the International Biometrics Society (IBS) Spring Meeting, Baltimore, Maryland, 03/2024.

2023:

- Invited Talk, 16th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics), Berlin, Germany, 12/2023.
- Invited Talk, Online Causal Inference Seminar, 11/2023.
- Invited Talk (Virtual), Statistics Seminar, Northest Normal University (China), 11/2023.
- Invited Talk, Algebraic Economics Workshop, Institute for Mathematical and Statistical Innovation (IMSI), Chicago, Illinois, 11/2023.
- Invited Talk, Big Data and Machine Learning in Econometrics, Finance, and Statistics, Chicago, Illinois, 10/2023.
- Invited Talk, Seminar, Econometrics and Statistics Colloquium, University of Chicago Booth School of Business, Illinois, 10/2023.
- Invited Talk, Algebraic Statistics for Ecological and Biological Systems Workshop, Institute for Mathematical and Statistical Innovation (IMSI), Chicago, Illinois, 10/2023.
- Invited Talk, Joint Statistical Meetings (JSM), Toronto, Canada, 08/2023.
- Invited Talk, ICSA 2023 Applied Statistics Symposium, Ann Arbor, Michigan, 06/2023.
- Invited Talk, 2023 Berkeley-Columbia Meeting in Engineering and Statistics, 04/2023.

2022:

- Invited Talk, Statistics Seminar, Tsinghua University, 10/2022.
- Invited Talk, Seminar, Fordham Psychometrics Program, Fordham University, 11/2022.
- Invited Talk, Seminar, Department of Statistics, Texas A&M University, 09/2022.
- Invited Talk, Seminar, Department of Mathematical Sciences, University of Texas at Dallas, 09/2022.
- Invited Talk, Joint Statistical Meetings (JSM), Washington, D.C., 08/2022.
- Invited Talk, International Chinese Statistical Association (ICSA) 2022 Applied Statistics Symposium, Gainesville, Florida, 06/2022.
- Invited Talk, New Advances in Statistics and Data Science, Honolulu, Hawaii, 05/2022.
- Invited Talk (Virtual), Seminar, Department of Mathematics, The Hong Kong University of Science and Technology, 03/2022.

2021:

- Invited Talk (Virtual), Seminar, Department of Statistics Colloquium Series at University of Missouri, 11/2021.
- Invited Talk, Seminar, Econometrics Colloquium at Department of Economics, Columbia University, 10/2021.
- Invited Talk, Department of Statistics Student Seminar, Columbia University, 09/2021.
- Invited Talk (Virtual), IMS Invited Session on "Inference, Optimization, and Computation on Discrete Structures", Joint Statistical Meetings (JSM), 08/2021.
- Invited Talk (Virtual), Bernoulli-IMS 10th World Congress in Probability and Statistics, July 2021.
- Invited Talk (Virtual), ETH Young Data Science Researcher Seminar, 04/2021.

2020:

- Invited Talk (Virtual), Early Career Speaker as the Lawrence D. Brown Ph.D. Student Award Recipient, Bernoulli-IMS One World Symposium, 08/2020.
- Invited Talk, Department of Statistics, Pennsylvania State University, 02/2020.
- Invited Talk, Department of Statistics, Rutgers University, 02/2020.
- Invited Talk, Department of Statistics, University of Wisconsin-Madison, 02/2020.
- Invited Talk, Department of Statistics, Columbia University, 01/2020.
- Invited Talk, Department of Statistics and Data Science, Cornell University, 01/2020.
- Invited Talk, Department of Mathematics and Statistics, McGill University, 01/2020.
- Invited Talk, Department of Statistics, University of California, Irvine, 01/2020.

- Invited Talk, Department of Statistics, Florida State University, 01/2020.
- Invited Talk, Department of Statistics and Operations Research, University of North Carolina at Chapel Hill, 01/2020.
- Invited Talk, Department of Data Sciences and Operations, Marshall School of Business, University of Southern California, 01/2020.
- Invited Talk, Department of Statistics, The Wharton School, University of Pennsylvania, 01/2020.
- Invited Talk, Department of Statistics, University of Washington, 01/2020.
- Invited Talk, Department of Statistics and Data Sciences, University of Texas at Austin, 01/2020.

2019:

- Invited Talk, School of Statistics, University of Minnesota, 12/2019.
- Invited Talk, Department of Statistics, University of Florida, 12/2019.
- Invited Talk, Department of Statistics, University of British Columbia, 12/2019.
- Invited Talk, Department of Statistics, Iowa State University, 12/2019.
- Invited Talk, Department of Statistics and Actuarial Science, University of Waterloo, 12/2019.
- Invited Talk, Department of Mathematics, University of Maryland, 11/2019.
- Invited Talk, Department of Statistics & Department of Psychology, University of Illinois at Urbana-Champaign, 11/2019.
- Invited Talk, International Chinese Statistical Association (ICSA) Applied Statistics Symposium, Raleigh, NC, 06/2019.

Other Talks and Presentations

- Talk in a Topic-Contributed Session, Joint Statistical Meetings (JSM) 2025, Nashville, Tennessee, 08/2025.
- Contributed Talk, International Meeting of the Psychometric Society (IMPS) 2025, Minneapolis, Minnesota, 07/2025.
- Talk at the "Statistics Meets Tensors" Workshop, IMSI, Chicago, 05/2025.
- Department of Statistics Student Seminar, Columbia University, 02/2025.
- Contributed Talk, International Meeting of the Psychometric Society (IMPS) 2023, University of Maryland, College Park, MD, 07/2023.
- Contributed Talk, International Meeting of the Psychometric Society (IMPS) 2021, Virtual, 07/2021.
- Contributed Talk, International Society for Bayesian Analysis 2021, Virtual, 06/2021.

- Research Talk, Michigan Institute for Data Science (MIDAS) 2019 Symposium, Ann Arbor, MI, 10/2019.
- Poster Presentation, Statistics in the Data Science Era: A Symposium to Celebrate 50 Years of Statistics at the University of Michigan, Ann Arbor, MI, 09/2019.
- Contributed Talk, Joint Statistical Meetings (JSM), Denver, CO, 07/2019.
- Poster Presentation, Midwest Machine Learning Symposium, Madison, WI, 06/2019.
- Contributed Talk, Joint Statistical Meetings (JSM), Vancouver, BC, Canada, 08/2018.
- Contributed Talk, International Meeting of the Psychometric Society (IMPS) 2018, New York, NY, 07/2018.
- Speed Oral and Poster Presentation, Michigan Student Symposium for Interdisciplinary Statistical Sciences (MSSISS), University of Michigan, Ann Arbor, MI, 04/2018.

Professional Services

Research Workshop Organizer:

- Organizer of the 2026 "*Statistical Foundations for Latent Structure and Representation Learning*" Workshop, Department of Statistics at Columbia University, November 4 – November 5, 2026.
- Organizer of the 2026 "*Workshop on Statistical Network Analysis and Beyond (SNAB)*", Vietnam, June 8 – June 10, 2026.
  - Co-organized with Can Le (UC Davis), Xiaodong Li (UC Davis), Jiashun Jin (Carnegie Mellon University), and Ji Zhu (University of Michigan).
- Organizer of the workshop "*Statistics meets Tensors: Methodology, Theory, and Applications*" at the Institute of Mathematics and Statistical Innovation, Chicago, May 5 – May 9, 2025.
  - Co-organized with Anru Zhang (Duke University), Yuxin Chen (University of Pennsylvania), and Cong Ma (University of Chicago).
- Organizer of the "*Psychometrics Workshop*" at Columbia University, September 22 – September 23, 2023.
  - Co-organized with Jingchen Liu and Zhiliang Ying (Columbia University).

Grant Proposal Reviewer and Panelist:

- National Science Foundation Panelist and Reviewer for the Statistics Program: 2024.
- National Science Foundation external reviewer for grant proposals: 2021, 2024.
- Reviewer for Columbia University Data Science Institute seed fund proposals: 2023.

Journal Editorial Board:

- Editorial Board Member for *Journal of Educational and Behavioral Statistics*.

Journal Reviewer:

- *Annals of Applied Statistics*, *Applied Psychological Measurement*, *Bayesian Analysis*, *Bernoulli*, *Behavior Research Methods*, *Biometrical Journal*, *Biometrics*, *Biometrika*, *British Journal of Mathematical and Statistical Psychology*, *Electronic Journal of Statistics*, *IEEE Transactions on Signal Processing*, *Journal of Business & Economic Statistics*, *Journal of Computational and Graphical Statistics*, *Journal of Educational and Behavioral Statistics*, *Journal of Machine Learning Research*, *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society: Series A*, *Journal of the Royal Statistical Society: Series B*, *Manufacturing and Service Operations Management*, *Multivariate Behavioral Research*, *Proceedings of the National Academy of Sciences*, *Psychometrika*, *Stat*, *Statistics in Medicine*, *Statistica Sinica*, *Technometrics*.

Paper Award Committee Member for:

- IMS Lawrence D. Brown Ph.D. Student Award: May 2023 – August 2026.
- IMS Thelma and Marvin Zelen Emerging Women Leaders in Data Science Award: Since 2024.
- American Statistical Association Statistical Learning and Data Science (SLDS) Student Paper Award: 2023, 2024, 2025.
- ASA Section on Bayesian Statistical Science (SBSS) Student Paper Competition: 2025.
- ASA Business and Economic Statistics Section Student Paper Competition: 2025.
- The 34th New England Statistics Symposium (NESS): September 2021.

IMS representative for AIC (African International Conference) 2027

IMS Representative and Program Committee Member for JSM 2026: IMS contributed session program chair.

Conference Session Organizer for:

- Invited Session, Statistical Learning and Data Science Conference, New York, 11/2026.
- Invited Session, Joint Statistical Meetings (JSM) 2026, Boston, Massachusetts.
- Topic-Contributed Sessions, JSM 2025, Nashville, Tennessee.
- Invited Sessions, International Chinese Statistical Association (ICSA) 2023 Applied Statistics Symposium, Ann Arbor, Michigan.
- Invited Session, International Society for Bayesian Analysis (ISBA) 2022, "Bayesian methods for complex dependent data with application to education and psychology".
- Invited Session, International Chinese Statistical Association (ICSA) 2022 Applied Statistics Symposium, "Latent Variable Models in the Data Science Era", Gainesville, Florida.

Local Organizing Committee Member: 2025 Lifetime Data Science Conference.

Internal Services at Columbia University

- Statistics Ph.D. Admission Committee: Academic years 2022–2023, 2024–2025, 2025–2026.
- Statistics Ph.D. Qualifying Exam Committee: 2023, 2024, 2025, 2026.
- Term Assistant Professor Search Committee: 2026.
- Statistics Seminar Organizing Committee: Academic years 2021–2022, 2022–2023, 2023–2024.
- Co-organizer, moderator, and panelist of the Graduate Research Opportunities Workshop "GROW Columbia 2024", Columbia University.
- Departmental Founder's Postdoc Recruitment Committee: 2024.
- First-year faculty mentors for Ph.D. students: Claire He (2022-2023), Chengzhu Huang (2023-2024), Nikira Walter (2024-2025), Wenjin Zhang (2025-2026).
- Statistics Ph.D. Dissertation Defense Committee: Seunghyun Lee (April 2026), Fangyi Chen (April 2026), Luhuan Wu (September 2025), Ye Tian (May 2025), Ling Chen (April 2025), Diane Lu (December 2023), Owen Ward (June 2022).
- Statistics Ph.D. Oral Exam (Dissertation Prospectus) Committee: Yi Han (September 2026), Chengzhu Huang (April 2026), David Snider (April 2026), Irene Chang (April 2026), Claire He (April 2026), Luhuan Wu (June 2024), Fangyi Chen (May 2024), Ling Chen (April 2024), Seunghyun Lee (September 2023), Ye Tian (May 2022).
- Econometrics Prospectus Defense Committee: Ruoqing Jiang, May 2023.
- Faculty Advisor for Master's Students in Applied Statistics (about 15 students in each cohort): Academic years 2021–2022, 2022–2023, 2023–2024, 2024–2025, 2025–2026.
- Guest Faculty Speaker: Academic Job Search Club, Department of Statistics, Columbia University, October 2021.

Professional Society Memberships

American Statistical Association

Institute of Mathematical Statistics

International Chinese Statistical Association

Psychometric Society